

templates\learn\what-is-cve.html

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{% extends "base.html" %}
{% block content %}
<!-- Main container with centered layout and responsive max-width -->
<div style="max-width:1400px; margin:0 auto 32px auto; padding-top:20px;">
<!-- Page Title & Intro -->
<!-- Peach color scheme for the main heading -->
<h1 style="color:#E2C2A2;">What is a CVE?</h1>
<!-- Comprehensive introduction explaining CVE purpose and importance -->
<p style="color:#E9ECE8; font-size:1.12em;">
<b>CVE</b> stands for
<span style="color:#D1AD87;"> Common Vulnerabilities and Exposures</span>.
It is the universal system managed by
<!-- Color-coded organization names for visual distinction -->
<span style="color:#D1AD87;">MITRE</span> and
<span style="color:#D1AD87;">NVD</span> for giving unique names to
publicly known cybersecurity vulnerabilities.
<br><br>
<span style="color:#D1AD87;"> <b>Why do CVEs matter?</b></span>
They allow security teams, researchers, and companies to discuss and fix the <u>same</u>
problem consistently using a single ID.
Every CVE describes a specific security flaw, explains its risk, and helps you learn
about real-world threats and how to manage them.
</p>
<!-- CVE Anatomy Interactive Sample -->
<!-- Educational section showing real CVE structure -->
<h2 style="color:#A8C7B5; margin-top:36px;">CVE Anatomy: What's Inside a Vulnerability
Record</h2>
<!-- Container with sage left border to emphasize importance -->
<div id="cve-sample-section"
style="background:#232b45; border-radius:16px; padding:34px 28px; margin-bottom:26px;
border-left:6px solid #A8C7B5;">
<!-- Flexible layout for CVE metadata fields -->
<div style="display:flex; flex-wrap:wrap; gap:22px; align-items:center;">
<!-- Interactive CVE ID field - uses famous Log4Shell as example -->
<span class="field-label" data-field="id" tabindex="0"
style="background:#263159;padding:7px
15px;border-radius:8px;color:#BFDAC8;cursor:pointer;font-family:monospace;">
ID: <b>CVE-2021-44228</b>
</span>
<!-- Publication date field with hover functionality -->
<span class="field-label" data-field="published" tabindex="0"
style="background:#263159;padding:7px 15px;border-radius:8px; color:#E9ECE8;
cursor:pointer;">
Published: <b>2021-12-09</b>
</span>
<!-- Critical severity with red background for urgency -->
<span class="field-label" data-field="severity" tabindex="0"
style="background:#f55855;padding:7px 15px;border-radius:8px; color:#fff;
font-weight:600; cursor:pointer;">
Severity: Critical
</span>
<!-- CWE classification with blue background -->
<span class="field-label" data-field="cwe" tabindex="0"
style="background:#3b8ded;padding:7px 15px;border-radius:8px; color:#fff;
font-weight:600; cursor:pointer;">
CWE: CWE-502
</span>
</div>
</div>
<!-- Description field spans full width -->
<div style="margin-top:18px;">
<span class="field-label" data-field="desc" tabindex="0"
style="background:#263159;padding:7px
15px;border-radius:8px;color:#E9ECE8;cursor:pointer;">
Description: Apache Log4j2 Remote code execution via JNDI lookup ("Log4Shell")
</span>
</div>
<!-- Reference links section -->
<div style="margin-top:13px; color:#a9adc1;">
<span class="field-label" data-field="refs" tabindex="0"
style="background:#263159;padding:5px 11px;border-radius:8px;cursor:pointer;">
References:
</span>
</div>
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<!-- External links to authoritative sources -->
<a href="https://nvd.nist.gov/vuln/detail/CVE-2021-44228"
target="_blank"
style="color:#BFDAC8; margin-left:10px;">
NVD
</a>
<a href="https://logging.apache.org/log4j/2.x/security.html" target="_blank"
style="color:#BFDAC8; margin-left:18px;">
Vendor Advisory
</a>
</div>
<!-- Tooltip Box - Hidden by default, positioned absolutely -->
<div id="cve-sample-tooltip" style="display:none; position:absolute; background:#101828;
color:#E9ECE8; border-radius:8px; padding:13px 15px; font-size:1.01em; min-width:210px;
max-width:350px; box-shadow:0 3px 18px #0008; z-index:200; pointer-events:none;"></div>
</div>
<!-- Section: How to Read & Interpret a CVE -->
<h2 style="color:#A8C7B5;">How to Read a CVE Entry & Interpret Its Details</h2>
<!-- Flexible layout with main content and sidebar tips -->
<div style="display:flex; flex-wrap:wrap; gap:30px; align-items:flex-start;">
<!-- Main content area with step-by-step guide -->
<ul style="background:#1a2236; border-radius:13px; padding:26px 28px 16px 26px;
color:#E9ECE8; font-size:1.09em; margin-bottom:22px;flex:2;">
<li><b> ID:</b> Unique reference for searching in vulnerability databases and
security tools.</li>
<li><b> Description:</b> Review the summary?does it match your environment and
risks?</li>
<li><b> Severity & Score:</b> The higher (e.g., <b>Critical</b>), the more urgent.
Check CVSS number if present.</li>
<li><b> CWE:</b> Type or category of bug?helps guide fixes and mitigations. E.g.,
Deserialization, XSS</li>
<li><b> References:</b> Advisories?ALWAYS consult for patches or step-by-step
guidance.</li>
<li><b> Products:</b> (If present) Specific affected software; always check
compatibility before action.</li>
</ul>
<!-- Sidebar with helpful tips and related links -->
<div style="flex:1; min-width:210px;">
<!-- Green accent color for tips section -->
<div style="background:#232b45; border-radius:13px; padding:18px 19px 19px 20px;
color:#E2C2A2; font-size:0.99em;">
<b>Tips:</b>
<!-- Practical advice list -->
<ul style="font-size:0.99em; color:#E9ECE8; margin-top:7px;">
<li>Never treat all CVEs the same?focus on critical/high scores first.</li>
<li>Check advisory links before taking action.</li>
<li>Look up the CWE type ("What is a CWE?") to prevent similar issues in your own
code.</li>
</ul>
<!-- Related learning links using Flask url_for routing -->
<div style="margin-top:11px;">
<a href="{{ url_for('learn.learn_topic', topic='what-is-cwe') }}" style="color:#BFDAC8;
text-decoration:underline;">What is a CWE?</a> |
<a href="{{ url_for('learn.learn_topic', topic='cvss-scores') }}" style="color:#BFDAC8;
text-decoration:underline;">CVSS Score Guide</a>
</div>
</div>
</div>
</div>
<!-- Interactive Today's CVEs -->
<h2 style="color:#A8C7B5; margin-top:38px;">Today's CVEs</h2>
<!-- Collapsible section for recent vulnerabilities -->
<div style="background:#1a2236; border-radius:14px; padding:20px 28px 24px 26px;
margin-bottom:18px;">
<!-- Header with toggle functionality and navigation link -->
<div style="display: flex; justify-content: space-between; align-items: center;
margin-bottom:9px; cursor: pointer;" onclick="toggleTodaysCVEs()">
<div style="color:#94a3b8;font-size:1.01em;">
<span>Showing most recent vulnerabilities published in the last 24 hours.</span>
<!-- Link to full vulnerabilities page -->
<a href="{{ url_for('main.vulnerabilities') }}" style="margin-left:17px;
color:#BFDAC8;text-decoration:underline;font-weight:600;">SHOW ALL TODAY CVEs ?</a>
</div>
<!-- Collapsible indicator arrow -->
<span id="todaysCvesIcon" style="color: #a9adc1; font-size: 1.2rem;">?</span>
</div>
<!-- Content area that can be toggled -->

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<div id="todaysCvesContent" style="display: block;">
<div style="display: grid; gap: 17px;">
<!-- Jinja2 logic to filter today's CVEs -->
{% set todays_cves = [] %}
{% for cve in latest_cves[:10] %}
<!-- Extract date from CVE published field -->
{% set date_str = cve.get('Published', '')[:10] %}
<!-- Compare with current date -->
{% if date_str == (now.strftime('%Y-%m-%d')) %}
<!-- Add to today's list if dates match -->
{% set _ = todays_cves.append(cve) %}
{% endif %}
{% endfor %}
<!-- Conditional rendering based on data availability -->
{% if todays_cves %}
<!-- Display first 5 CVEs from today -->
{% for cve in todays_cves[:5] %}
<!-- CVE card with severity-based color coding -->
<div style="background:#263159; border-radius:12px; padding:18px 19px 14px 19px;
border-left:4px solid {% if cve.get('Severity') == 'CRITICAL' %}#f47174{% else
%}#8DAD9C{% endif %};">
<!-- Card header with CVE details -->
<div style="display:flex; justify-content:space-between; align-items:flex-start;
margin-bottom:10px;">
<div style="flex:1;">
<!-- CVE ID, severity badge, and CWE info -->
<div style="display:flex; gap:11px; align-items:center;">
<!-- Clickable CVE ID linking to detail page -->
<a href="{{ url_for('main.cve_detail', cve_id=cve.get('ID')) }}" style="color:#BFDAC8;
font-weight:700; font-size:1.1em; text-decoration:none;">
{{ cve.get('ID', 'N/A') }}
</a>
<!-- Color-coded severity badge -->
<span style="background: {% if cve.get('Severity')== 'CRITICAL' %}#f47174{% elif
cve.get('Severity')== 'HIGH' %}#fca311{% elif cve.get('Severity')== 'MEDIUM' %}#3498db{%
else %}#42d392{% endif %}; color:white; padding:4px 12px; border-radius:16px;
font-size:0.83em; font-weight:600;">
{{ cve.get('Severity', 'UNKNOWN') }}
</span>
<!-- Optional CWE classification -->
{% if cve.get('CWE') %}
<span style="margin-left:10px; background:#212d46; color:#BFDAC8; padding:1px 9px;
border-radius:11px; font-size:0.95em;">{{ cve.get('CWE') }}</span>
{% endif %}
</div>
<!-- Truncated description with ellipsis -->
<div style="color:#E9ECE8; line-height:1.44; margin-top:7px; font-size:0.97em;">
{{ cve.get('Description', 'No description available')[:170] }}{% if
cve.get('Description', '')|length > 170 %}...{% endif %}
</div>
</div>
<!-- Publication date aligned to right -->
<div style="min-width:105px; text-align:right; color:#a9adc1;">
<div>{{ cve.get('Published')[:10] }}</div>
</div>
</div>
</div>
{% endfor %}
{% else %}
<!-- Empty state when no CVEs found for today -->
<div style="text-align:center; color:#94a3b8; padding:26px;font-size:1.08em;">
<div style="font-size:1.7rem;margin-bottom:8px;">?</div>
No new CVEs detected today.
</div>
{% endif %}
</div>
</div>
</div>
<!-- Timeline of Major CVEs -->
<h2 style="color:#A8C7B5; margin-top:34px;">Timeline of Major CVEs</h2>
<!-- Historical context table -->
<div style="background:#1a2236;border-radius:10px;padding:24px 18px 20px 18px;
margin-bottom:18px;">
<!-- Structured table of famous CVEs -->
<table style="width:100%; color:#E9ECE8; font-size:1.05em; border-collapse:collapse;">
<thead>
<!-- Table headers -->

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Year	CVE ID	Nickname	Impact & Legacy
2014	CVE-2014-0160	Heartbleed	SSL bug exposed private keys, triggered global emergency patching
2014	CVE-2014-6271	Shellshock	Bash shell flaw affected millions of Linux servers
2017	CVE-2017-0144	EternalBlue	Used in WannaCry ransomware; showed the danger of wormable network bugs
2017	CVE-2017-5638	Apache Struts	Enabled the Equifax breach, crucial lessons for enterprise patching
2021	CVE-2021-44228	Log4Shell	Critical bug in Log4j2 affected cloud, apps, millions of devices worldwide
2022	CVE-2022-22965	Spring4Shell	New Java bug affected popular Spring framework (recent ecosystem risk)

Explanatory note about historical significance -->
 <div style="margin-top:14px; color:#a9adc1; font-size:0.97em;">
 Why These Matter:
 Each event reshaped industry patching, compliance, and risk management. Major spikes in awareness and new best practices followed these discoveries.
 </div>
 </div>
 </div>

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<!-- Interactive Highlight/Tooltip JS -->
<script>
// Tooltip data object containing educational information
const cveSampleTooltips = {
'id': { title: "CVE ID", text: "A unique vulnerability identifier. Used to track and
search. Format: CVE-YYYY-NNNNN." },
'published': { title: "Published Date", text: "Date the vulnerability was publicly
announced. Crucial for understanding risk exposure windows." },
'severity': { title: "Severity", text: "Shows risk level (Critical, High, Medium,
Low)?often based on the CVSS score. Prioritize higher severities." },
'cwe': { title: "CWE", text: "Specifies the underlying type of coding/design flaw. Links
to common patterns for prevention (e.g. Deserialization, XSS)."},
'desc': { title: "Description", text: "A summary of what the vulnerability is and how it
can be exploited. Always read to see if it applies to your environment." },
'refs': { title: "References", text: "Links to authoritative advisories, patches, vendor
updates, and analysis. The best source of detailed remediation steps." }
};
// Immediately invoked function expression to avoid global namespace pollution
(function() {
// Get DOM references for tooltip system
const section = document.getElementById('cve-sample-section');
const tooltip = document.getElementById('cve-sample-tooltip');
let active = null; // Track currently active tooltip
// Function to display tooltip with positioning logic
function showTip(label, event) {
const field = label.dataset.field; // Get field type from data attribute
const data = cveSampleTooltips[field]; // Retrieve tooltip content
if (!data) return; // Exit if no data found
// Set tooltip content with title and description
tooltip.innerHTML = "<b style='color:#E2C2A2;'>" + data.title + "</b><br>" + data.text;
tooltip.style.display = "block";
// Calculate optimal positioning
const rect = label.getBoundingClientRect();
tooltip.style.position = 'fixed';
let left = rect.left;
let top = rect.bottom + 8; // Position below the trigger element
// Prevent tooltip from going off-screen
if((left + tooltip.offsetWidth + 10) > window.innerWidth){
left = window.innerWidth - tooltip.offsetWidth - 22;
}
// Apply calculated position
tooltip.style.left = left + "px";
tooltip.style.top = top + "px";
active = label; // Set as currently active
}
// Function to hide tooltip and reset state
function hideTip(ev) {
tooltip.style.display = "none";
active = null;
}
// Set up event listeners for all interactive elements
const labels = section.querySelectorAll('.field-label');
labels.forEach(function(label) {
// Mouse events for desktop interaction
label.addEventListener('mouseover', function(e) { showTip(label, e); });
label.addEventListener('mouseleave', hideTip);
// Keyboard events for accessibility
label.addEventListener('focus', function(e){ showTip(label, e); });
label.addEventListener('blur', hideTip);
// Make elements focusable for keyboard navigation
label.setAttribute('tabindex', '0');
});
// Global event handlers for cleanup
document.addEventListener('mousedown', function(e){
// Hide tooltip when clicking elsewhere
if(active && !active.contains(e.target)) hideTip();
});
// Hide on scroll to prevent positioning issues
window.addEventListener('scroll', hideTip, {passive:true});
// Hide on resize to prevent positioning issues
window.addEventListener('resize', hideTip);
// Hide on escape key for accessibility
document.addEventListener('keydown', function(e){
if(e.key === 'Escape') hideTip();
});
})();
// Today's CVEs collapse function - standalone for onclick handler

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function toggleTodaysCVEs() {
// Get DOM references
const content = document.getElementById('todaysCvesContent');
const icon = document.getElementById('todaysCvesIcon');
// Toggle visibility and update icon
if (content.style.display === 'none') {
content.style.display = 'block';
icon.textContent = '?'; // Down arrow for expanded state
} else {
content.style.display = 'none';
icon.textContent = '?'; // Right arrow for collapsed state
}
}
</script>
<!-- Navigation back to main dashboard -->
<div style="text-align:center; margin:2.5em 0 1em 0;">
<a href="{{ url_for('main.index') }}" style="color:#BFDAC8; padding:0.8em 2em;
border-radius:6px; font-weight:600; text-decoration:none;">
? Back to Dashboard
</a>
</div>
{% endblock %}

```